Qasim Bin Saeed

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Professional Summary

Researcher with a background in Electrical Engineering and graduate research experience in robotics, computer vision, machine learning, and autonomous systems. My work focuses on robot perception, visual localization, autonomous navigation, and learning-based 3D scene reconstruction, with experience in Structure-from-Motion, neural scene representations, UAV autonomy, and ROS-based robotic systems. I am particularly interested in developing reliable perception and learning methods that enable robots to understand, navigate, and operate in complex real-world environments. My long-term research interests lie in robotics, embodied AI, 3D vision, and intelligent autonomous systems.

Core Skills

Robotics and Autonomous Systems: ROS 2, Gazebo, NVIDIA Isaac Sim, visual localization, visual SLAM, GPS-denied navigation, waypoint planning, sensor fusion, PID control, camera–IMU integration, UAV autonomy, quadruped robot navigation
Computer Vision and Deep Learning: PyTorch, TensorFlow, OpenCV, convolutional neural networks (CNNs), Transformers, Vision Transformers (ViTs), object detection, semantic segmentation, instance segmentation, feature extraction and matching, visual tracking
3D Vision and Robot Perception: Structure-from-Motion (SfM), Multi-View Stereo (MVS), multi-view geometry, camera calibration, camera pose estimation, Neural Radiance Fields (NeRF), 3D Gaussian Splatting (3DGS), neural rendering, point-cloud processing, 3D reconstruction
Programming and Computing: Python, C++, MATLAB, CUDA, NumPy, Pandas, Git, Docker, Linux
Simulation and Research Tools: Gazebo, NVIDIA Isaac Sim, Unreal Engine 5, 3D Slicer, MONAI

Experience

University of Manitoba **May 2024 – Present**
Teaching and Research Assistant *Winnipeg, MB, Canada*

- Conduct research on AI and metaverse-enabled structural health monitoring, building deep learning workflows for infrastructure 3D reconstruction, damage detection, and analysis in UAV- and VR-based environments.
- Design and iterate on machine learning experiments, prepare research datasets and feature pipelines, and evaluate model behavior across simulated and image-based structural inspection tasks.
- Work across multi-source inputs including image, simulation, and inspection data to support prediction workflows, comparative experiments, and clear measurement of model performance.
- Co-authored and first-authored research outputs for top-tier venues, including work under review or in preparation for *Automation in Construction*, *IEEE Transactions on Industrial Informatics*, and *Scientific Reports*.
- Presented technical work to research audiences and collaborated closely with faculty and interdisciplinary stakeholders on problem definition, methodology, and communication of results.
- Delivered tutorials and labs for CIVL 3590 and ENG 2040 covering numerical methods, MATLAB programming, and technical communication; graded assessments, supported office hours, and created supplemental learning resources.

Mitacs Globalink **Jun 2026 – Present**
Research Internship Mentor *Winnipeg, MB, Canada*

- Mentored ten Mitacs Globalink research interns from Colombia, Mexico, China, and Pakistan, supporting students with different academic backgrounds and research expectations.
- Provided practical research guidance, regular communication, and respectful support throughout the internship period.

Moffitt Cancer Center **Jun 2023 – Jan 2024**
Medical Imaging Research Assistant *Florida, USA*

- Built and improved medical image segmentation workflows using PyTorch and MONAI for multi-dimensional MRI and CT data in NIfTI and DICOM formats.
- Prepared training datasets and data pipelines through normalization, augmentation, cropping, annotation, and auto-labeling workflows in 3D Slicer and MONAI Label, strengthening dataset quality and model readiness.
- Developed and optimized a high-accuracy segmentation model to support tumor detection and analysis, contributing

to reproducible experimentation across preprocessing, training, evaluation, and iterative model improvement stages.
- Supported reliable research workflows by standardizing preprocessing and data preparation steps for repeated training runs and consistent downstream analysis.

Education

University of Manitoba **May 2024 – Aug 2026**
Master of Science (MSc) in Engineering *Winnipeg, MB, Canada*

- CGPA: 4.25/4.5
- Relevant coursework: Natural Language Processing, Deep Generative Modelling, Structural Health Monitoring, Advanced Deep Learning for Structural Health Monitoring, Data Science with Real-World Datasets
- Graduate work emphasizes machine learning experimentation, feature engineering, model evaluation, and applied data science on real-world datasets, including structured and tabular data analysis workflows.
- Awards: NSERC funding, Mitacs Lab2Market, Invention to Innovation(i2I), International Graduate Student Entrance Scholarship (IGSES), Mitacs Globalink Mentor

National University of Sciences and Technology (NUST) **Sept 2019 – Jun 2023**
Bachelor of Electrical Engineering *Islamabad, Pakistan*

- CGPA: 3.57/4.0
- Relevant coursework: Data Structures and Algorithms, Object-Oriented Programming, Linear Control Systems, Machine Learning, Computer Vision, Numerical Methods, Probability and Statistics, Digital Image Processing, Digital Signal Processing, Signals and Systems, Instrumentation and Measurements
- Awards: Three-time NUST High Achiever Award, Ehsaas Scholarship covering full four-year tuition

Publications and Research

1. **Bin-Saeed, Q.** (2026). PIRA: Learning Robust Diffusion Policies from Rescue Interventions. *IEEE Robotics and Automation Letters* (**IF: 5.3**). In preparation.
2. **Bin-Saeed, Q.**, and Cha, Y. J. (2026). Autonomous UAV-based advanced 3D reconstruction for Structural Health Monitoring of Civil Infrastructure. *Automation in Construction* (**IF: 12.6**). In preparation.
3. **Bin-Saeed, Q.**, and Cha, Y. J. (2026). Advances in 3D Reconstruction for Civil Engineering: From Structure-from-Motion to NeRF and 3D Gaussian Splatting. *Automation in Construction* (**IF: 12.6**). Under Review.
4. **Bin-Saeed, Q.**, and Cha, Y. J. (2026). Hybrid-NeRF: A Geometry- and Depth-Aware Gaussian Neural Radiance Field for Pixelwise 3D Damage Mapping. *ISPRS Journal of Photogrammetry and Remote Sensing* (**IF: 12.9**). Under Review.
5. Ali, R., **Bin-Saeed, Q.**, Buyukozturk, O., Lee, S., and Cha, Y. J. (2026). Monocular computer vision-based simultaneous pothole segmentation and 3D volume prediction using 3DPredictNet. *IEEE Transactions on Industrial Informatics* (**IF: 10.4**). DOI: 10.1109/TII.2026.3706348.
6. **Bin-Saeed, Q.**, and Cha, Y. J. (2025). Multi-modal deep-attention-BiLSTM based early detection of mental health issues using social media posts. *Scientific Reports* (**IF: 4.3**), 15, 35152. DOI: 10.1038/s41598-025-19141-0.

- **Conference presentations**: Engineering Mechanics Institute Conference (UC Boulder, 2026), Engineering Mechanics Institute Conference (UC Irvine, 2025), and International Workshop on Structural Health Monitoring (Stanford University, 2025).

Academic Reviewer Service

- ASCE Journal of Computing in Civil Engineering, (Q1 | IF: 5.8)
- IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2026, (CORE Rank B)

Community Service

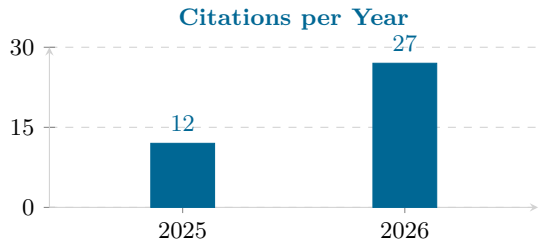
- Volunteered at the 2025 Manitoba Student Health Research Forum, the 2024 and 2025 CEGSA Graduate Research Competitions, and the 2025 CSCE Conference in Winnipeg.
- Participated in and won the 2025 CEGSA Graduate Research Competition.
- Participated in the CIBC Run for the Cure and registered as a volunteer with Bear Clan Patrol in Winnipeg.
- Served as Associate Director and Deputy Director of the NUST Volunteer Club, coordinating student-led community service initiatives.

- Contributed to the Chadar Street Store initiative, through which our team distributed more than 4,000 donated items to over 700 families.
- Mentored junior engineering students in programming and provided academic and technical guidance.

Research Impact

Metric	All	Since 2021
Citations	39	39
h-index	3	3
i10-index	2	2

Self-Citation Rate: **0.0%** | Source: Google Scholar



References

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Dr. Rahamat Ali
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